

A Hilbert-Pair Subcase For Norm-Attaining Projective Tensors

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Abstract

Dantas, Jung, Roldan, and Rueda Zoca ask whether, for reflexive Banach spaces X and Y , norm-attaining tensors are dense in $X\widehat{\otimes}_\pi Y$. They prove that every tensor in $H\widehat{\otimes}_\pi H$ attains its projective norm when H is a complex Hilbert space. We record the same conclusion for arbitrary pairs of complex Hilbert spaces H_1, H_2 . Consequently, every nuclear operator between two complex Hilbert spaces attains its nuclear norm. The result also extends to finite ℓ_1 -sums of Hilbert spaces in the first variable and a Hilbert space in the second variable. This is a positive subcase, not a solution of the general reflexive problem.

1 Source Question

The source paper introduces norm-attainment for projective tensors and nuclear operators. In Section 6, Question 6.1 asks:

Let X, Y be reflexive Banach spaces. Is it true that the set of all norm-attaining tensors is dense in $X\widehat{\otimes}_\pi Y$?

The same section recalls two positive cases already proved in the paper: every tensor in $H\widehat{\otimes}_\pi H$ attains its projective norm for a complex Hilbert space H , and norm-attaining tensors are dense in $L_p(\mu)\widehat{\otimes}_\pi L_q(\nu)$ for $1 < p, q < \infty$.

2 Definitions

Let X and Y be Banach spaces. A tensor

$$z = \sum_{n=1}^{\infty} \lambda_n x_n \otimes y_n, \quad \lambda_n > 0, \quad x_n \in S_X, \quad y_n \in S_Y,$$

is said to attain its projective norm if

$$\|z\|_\pi = \sum_{n=1}^{\infty} \lambda_n.$$

Equivalently, the projective norm of z is realized by an absolutely summable representation by simple tensors.

3 Idea Of The Proof

For Hilbert spaces, projective tensors are the same object as trace-class operators. A nuclear operator $T : H_1 \rightarrow H_2$ has a singular-value decomposition

$$T = \sum_n s_n \langle \cdot, u_n \rangle v_n,$$

where (u_n) and (v_n) are orthonormal systems and $\sum_n s_n$ is the trace-class, hence nuclear, norm. Reading the same formula as a tensor representation gives an optimal projective representation.

For finite ℓ_1 -sums, the projective tensor product distributes over the ℓ_1 -sum. Thus the norm of a tensor is the sum of the norms of its Hilbert components, and concatenating the singular-value representations of the components gives a single optimal representation.

4 Hilbert Pairs

Theorem 1. *Let H_1 and H_2 be complex Hilbert spaces. Then every tensor in $H_1 \widehat{\otimes}_\pi H_2$ attains its projective norm.*

Proof. Since Hilbert spaces have the approximation property, the canonical map

$$H_1 \widehat{\otimes}_\pi H_2 \longrightarrow \mathcal{N}(H_1^*, H_2)$$

is an isometric isomorphism, where $h \otimes k$ acts on H_1^* by $\varphi \mapsto \varphi(h)k$. Thus it is enough to prove that every nuclear operator $T : H_1^* \rightarrow H_2$ attains its nuclear norm.

Let $T : H_1^* \rightarrow H_2$ be nuclear. Since H_1^* is again a Hilbert space, nuclear operators are exactly trace-class operators. Hence T has a singular-value decomposition

$$T\varphi = \sum_{n=1}^N s_n \psi_n(\varphi) v_n,$$

where $N \in \mathbb{N} \cup \{\infty\}$, $s_n > 0$, (ψ_n) is an orthonormal system in $(H_1^*)^*$, (v_n) is an orthonormal system in H_2 , and

$$\|T\|_{\mathcal{N}} = \sum_{n=1}^N s_n.$$

By reflexivity of H_1 , each $\psi_n \in (H_1^*)^*$ is evaluation at a unique unit vector $u_n \in H_1$. Under the tensor–nuclear identification, this is precisely the tensor representation

$$z_T = \sum_{n=1}^N s_n u_n \otimes v_n,$$

with all vectors of norm one and with coefficient sum equal to the projective norm. Therefore z_T , and hence every element of $H_1 \widehat{\otimes}_\pi H_2$, attains its projective norm. $\square$

Corollary 1. *Let H_1, H_2 be complex Hilbert spaces. Then every nuclear operator $T : H_1 \rightarrow H_2$ attains its nuclear norm.*

Proof. This is the operator form of the proof of Theorem 1. $\square$

5 Finite ℓ_1 -Sums Of Hilbert Spaces

Theorem 2. *Let $H_1, \dots, H_m$ and K be complex Hilbert spaces, and put*

$$X = H_1 \oplus_1 \cdots \oplus_1 H_m.$$

Then every tensor in $X \widehat{\otimes}_\pi K$ attains its projective norm.

Proof. The projective tensor product commutes isometrically with finite ℓ_1 -sums:

$$X \widehat{\otimes}_\pi K \cong (H_1 \widehat{\otimes}_\pi K) \oplus_1 \cdots \oplus_1 (H_m \widehat{\otimes}_\pi K).$$

Indeed, the dual of the left-hand side is $\mathcal{L}(X, K^*)$, which is isometric to $\mathcal{L}(H_1, K^*) \oplus_\infty \cdots \oplus_\infty \mathcal{L}(H_m, K^*)$, the dual of the right-hand side.

Let $z \in X \widehat{\otimes}_\pi K$, and write

$$z = (z_1, \dots, z_m), \quad z_j \in H_j \widehat{\otimes}_\pi K.$$

By Theorem 1, each nonzero z_j admits a representation

$$z_j = \sum_{n=1}^{N_j} s_{j,n} h_{j,n} \otimes k_{j,n}, \quad \sum_{n=1}^{N_j} s_{j,n} = \|z_j\|_\pi,$$

with $s_{j,n} > 0$, $h_{j,n} \in S_{H_j}$, and $k_{j,n} \in S_K$. Embedding H_j into the j -th coordinate of X and concatenating these representations gives

$$z = \sum_{j=1}^m \sum_{n=1}^{N_j} s_{j,n} \iota_j(h_{j,n}) \otimes k_{j,n}.$$

The coefficient sum is

$$\sum_{j=1}^m \sum_{n=1}^{N_j} s_{j,n} = \sum_{j=1}^m \|z_j\|_\pi = \|z\|_\pi,$$

where the last equality is the ℓ_1 -sum norm. Hence z attains its projective norm. $\square$

6 Scope And Limitations

Finite ℓ_1 -sums of Hilbert spaces are reflexive, so Theorem 2 is a positive subcase of Question 6.1. The result is still narrow: it does not address arbitrary reflexive spaces, and it gives no counterexample or full criterion for density of norm-attaining tensors.

The argument also does not solve the source paper's Question 6.2 about whether there are Banach spaces X, Y for which norm-attaining nuclear operators are not dense in $\mathcal{N}(X, Y)$, nor Question 6.3 concerning property α or quasi- α .

7 Verification Notes

The proof uses only standard Hilbert-space trace-class theory and the standard isometric identification between projective tensors and nuclear operators when the relevant Hilbert space has the

approximation property. The source paper’s Proposition 3.8 is the diagonal case $H_1 = H_2$; Theorem 1 is the same singular-value proof without requiring the domain and range Hilbert spaces to coincide.

Cheap run indexes were searched for prior packets on arXiv:2006.09871 and for the relevant norm-attaining tensor/nuclear-operator keywords. Bounded web searches found the source paper but no later exact answer to Question 6.1. Novelty confidence should be considered modest because this is an elementary extension of a standard trace-class argument.

8 References

- S. Dantas, M. Jung, O. Roldan, and A. Rueda Zoca, *Norm-attaining tensors and nuclear operators*, arXiv:2006.09871.
- I. Gohberg, S. Goldberg, and M. A. Kaashoek, *Classes of Linear Operators*, Vol. I, cited in the source paper for the singular-value representation and trace-class norm formula.
- R. A. Ryan, *Introduction to Tensor Products of Banach Spaces*, cited in the source paper for the tensor/nuclear identifications and projective tensor basics.